

DSP FOR “MAKE IN INDIA”

EE 338 Application Assignment

Group 1

APPLICATIONS OF DSP

- A significant chunk of modern technology relies on signal processing (especially digital signal processing).
- DSP is widely used in medicine to extract exact features of the imaged object. As an example, MRI scans are often processed to get clearer images of whatever organ needs to be observed.
- The audio industry heavily relies on signal processing for applications like file compression and active noise cancellation.
- DSP finds uses in defense and geographical analysis. Satellite images are often processed to obtain distinguishing features of the terrain and discern natural features, structures built by the enemy, etc.

MAKE IN INDIA

- We take this opportunity to introduce two topics that we have worked on : Region of Interest Extraction and Motion De-Blur. We feel that they can have applications for real-life problems.
- Region of interest extraction is widely needed for medical equipment and terrain mapping. We believe a fast implementation of the same can go a long way in helping design high speed and high efficiency devices for defense and medical purposes within India.
- Motion de-blur has much more scope for commercial applications. A proposed device we could design based on this would be a DSLR that removes any blur effects (often a problem with fast moving image subjects). Additionally, it also finds uses in the defense industry.



MOTION DE-BLUR

Our primary topic

UNDERLYING THEORY

- At the heart of it all is the Wiener filter, which produces an estimate of the original image by minimising the expected mean square error between the true and estimated images.
- It minimizes the blow-up of noise during deconvolution, especially at frequencies where the signal to noise ratio is very poor.

$$L(u, v) = \frac{H^*(u, v)}{\frac{S_n(u, v)}{S_f(u, v)} + |H(u, v)|^2}$$

- $L(u, v)$: Wiener Filter
- $H(u, v)$: motion model Fourier Transform
- $S_n(u, v)$: Powerspectrum of Noise
- $S_f(u, v)$: Powerspectrum of signal

ASSUMPTIONS

- Noise is independent of the image
- Noise statistics do not change spatially
- Either the image or the noise (or both) has (have) zero mean
- Second order statistics (i.e. power spectrum) of the image and the noise are known

RESULTS



Original Image



Motion blurred
Image



Output Image

RESULTS



Original Image



Motion blurred
Image



Output Image

RESULTS



Original Image



Motion blurred
Image



Output Image

FURTHER APPLICATIONS

- In high speed traffic it is very difficult for an ordinary surveillance camera to take a sharp image of moving cars. Weiner filter can be very effective in reading the number plate of the moving car. This can be a cheap alternative for surveillance in the roads as no additional hardware is required
- Weiner filters implementation can also be used to turn a low budget camera into an motion camera, which can indeed be used for wildlife photography in which capturing a still image is must.



REGION OF INTEREST EXTRACTION

An additional application

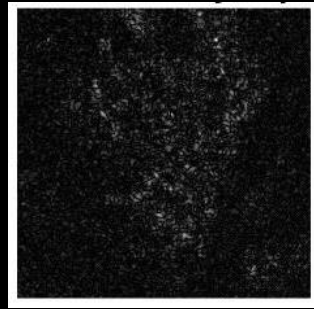
IMPLEMENTATION

- We implement the paper "Region-of-Interest Extraction Based on Frequency Domain Analysis" by Zhang and Yang for this particular application. We chose this over other techniques as it proposes a faster and more efficient extraction compared to traditional techniques.
- We work on the HSI colour space. Additionally, we exploit the quaternion Fourier transform to extract the saliency map of the image.
- Once we've obtained the saliency map, our task simplifies to applying a Gaussian pyramid over the map, properly thresholding it and multiplying it (in the frequency domain) with our original image to get the final output image (in the frequency domain).
- Our results, while not exactly as good as the paper, have come close and show good results when we tune the parameters correctly.

RESULTS OBTAINED



Original Image



Initial saliency map



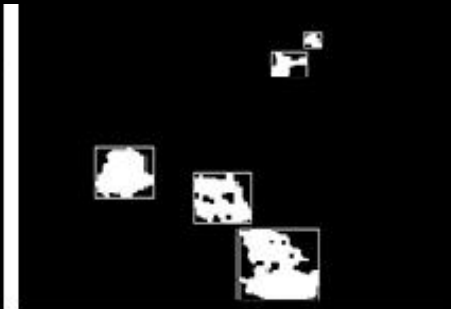
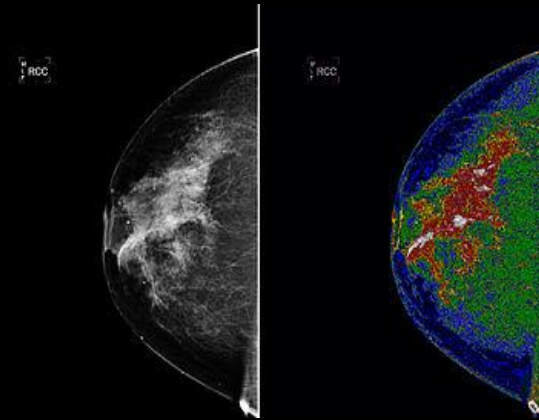
Final saliency map



Final Image

FURTHER APPLICATIONS

- Shown alongside are some applications of this method, including vehicle detection from roads and medical analysis.



(a)Original image

(b) Method 4

(c) Proposed method

Fig. 9. Result analysis



SOME OBSERVATIONS

- As can be observed, the algorithm performs fairly well. However, the results are probably not as fool proof or consistent as we would like to see.
- A lot of time was spent tuning the parameters as well to get perfect results for every image. This probably could have been reduced by a better implementation.
- Considering the above points, we decided to take the opportunity to investigate into another topic that we felt could be used for providing real life applications using DSP.